
(a) Speed ripple per setting
Speed ripple, std at steady state (RPM)
Setting (dashed line = baseline)
(b) Speed ripple at 1× and 2× shaft frequency
Speed-ripple amplitude (RPM, zero-to-peak)
1× shaft frequency
2× shaft frequency
0.29
0.36
A
Setting (dashed lines = baseline)
(c) Flatter PWM vs steadier speed
Speed ripple, std (RPM) ↓ steadier speed
PWM jitter (% duty) ← flatter command